3 Computational thinking: Catch the mouse



You will learn

- to use the mouse to control the computer
- to use the keyboard to control the computer
- what inputs are
- to see and hear outputs.

In this unit you will use a computer. You will use the computer mouse. You will make a cat move on the screen.



Class activity

A computer **mouse** looks like this. You can move it on the table.

Have you used a computer mouse before?

Draw a picture to show how you used the mouse.



Learning outcomes: Run and use a simple program made by someone else; Describe a program by saying what its inputs and outputs are



input output
Scratch game
mouse pointer



Talk about...

Some computers don't have a mouse. You touch the screen instead. Have you ever done that?



Did you know?

Doug Engelbart invented the computer mouse. His friends asked why he called it a mouse. He said, "Because it looks like a little mouse with a tail." Does your mouse have a tail or not?



3.1 Hello Scratch

In this lesson

You will learn:

- about **Scratch**. Scratch is a programming language for children.



This is Scratch the cat. He will help you learn.

Scratch will walk when you click with the mouse. Scratch walks inside a square. The square is called the **stage**.

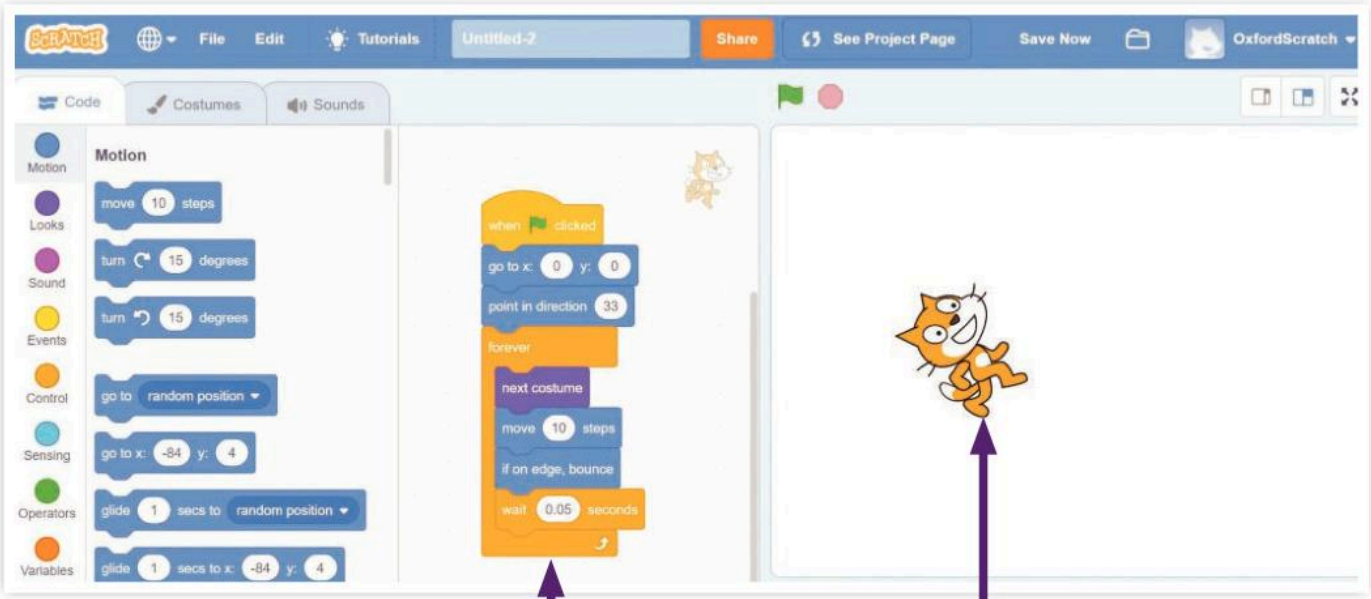


Activity

You will work with the Scratch website.

https://scratch.mit.edu/projects/editor/

A program is ready to use. The program tells Scratch to walk.



This is the program.

This is Scratch.

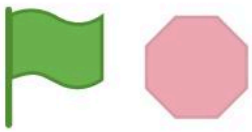
Icons are little pictures. They show you things the computer can do.

To use an icon do this:

1 Move the mouse pointer to the icon.

2 Click the icon.

There are two icons at the top of the stage. They look like this.



The green flag makes the program start. The red button makes the program stop.

1 Move the mouse pointer to the green flag. Click on the green flag to make Scratch walk.



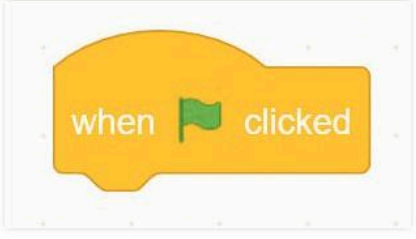
2 Move the mouse pointer to the red button. Click on the red button to make Scratch stop.



Think again

Look at the program. The program is made of blocks.

What is the picture on the first block? Why do you think the block shows that picture?



3.2 Scratch on stage

In this lesson

You will learn:

➔ how to make changes to the Scratch stage.

Last lesson you used the 'start' and 'stop' icons. By clicking the icons you could start and stop the program. Scratch walked on the stage. In this lesson you will make changes to the stage.



You can change the size of the stage.

This icon makes the stage big. Move the pointer to the icon. Click the mouse button.



This icon makes the stage small.



Click on this icon. Make the stage small again.



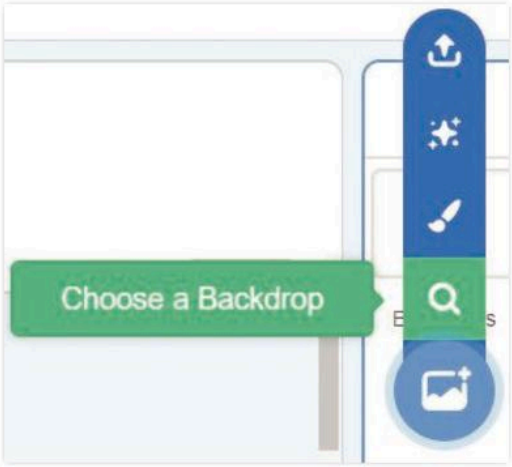
You can change the **backdrop**.

The stage can have a picture on it. The picture is called the backdrop.

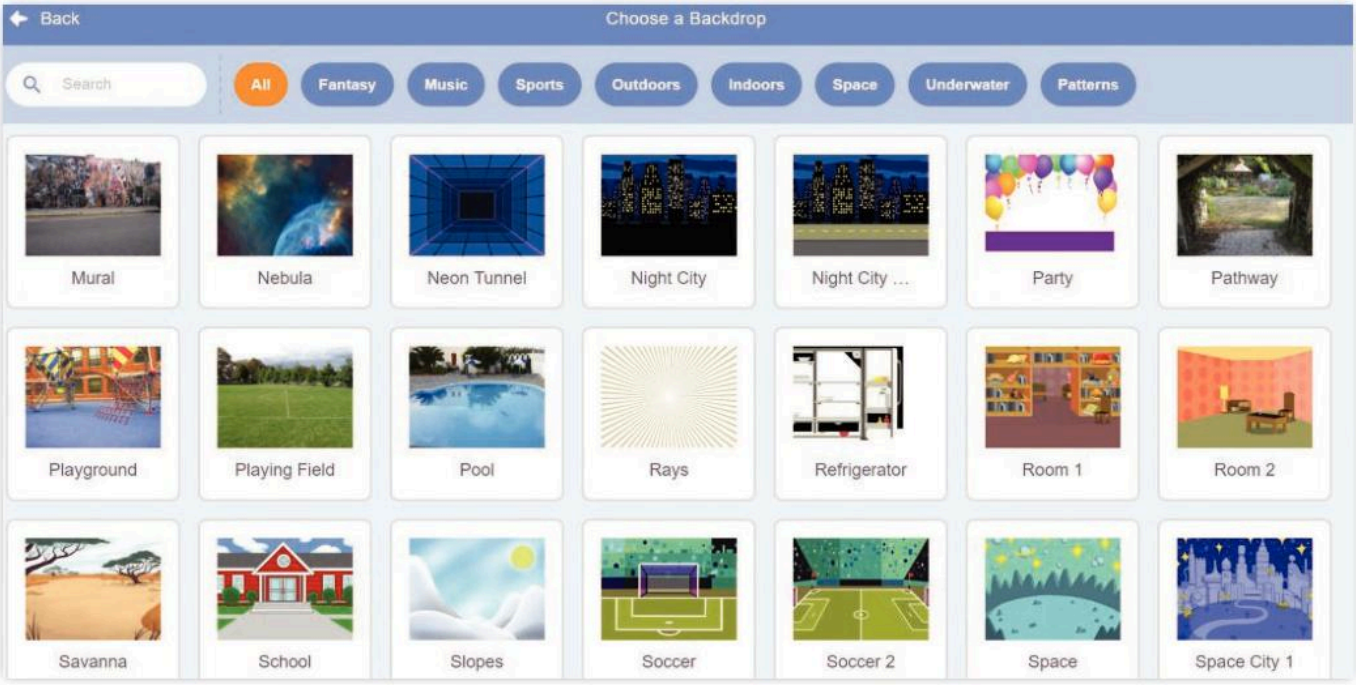


This icon lets you change the backdrop.

Click here to choose a new backdrop.



Click on any picture.



What does the stage look like now?

Be creative

Think of a new backdrop for Scratch. Paint a picture to show your idea.

3.3 Chase the mouse

In this lesson

You will learn:

➔ about input.

Input is how you control the computer. You can use a:

- mouse
- keyboard.

Mouse

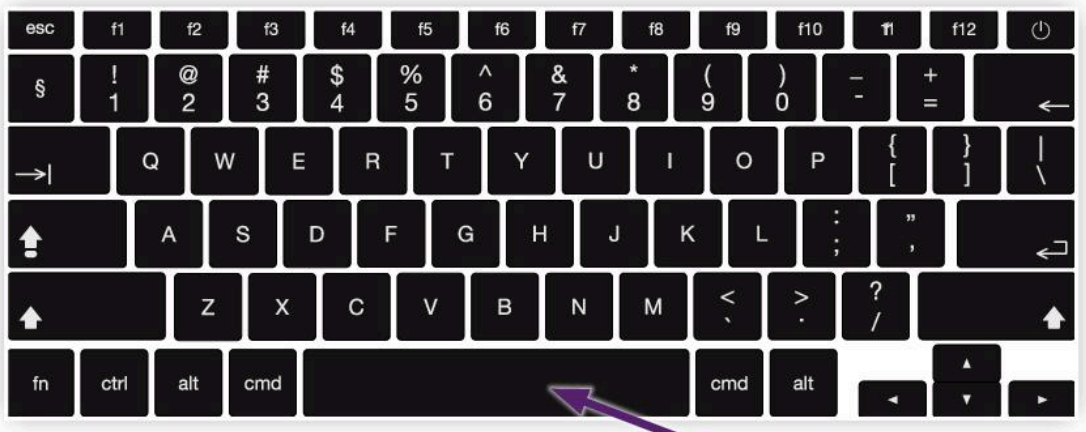
The **mouse pointer** is on the screen.

Move the mouse. The mouse pointer will move too.



Keyboard

Your computer has a **keyboard**.



The keyboard lets you type letters.

Activity

You can move the mouse. Press the space bar. Scratch will start to walk.

Move the mouse pointer. Scratch will chase after it.



Keep moving the mouse pointer.
Don't let Scratch catch it.

Think again

You have learned two ways to control the computer.

Draw or write to show these two ways.



3.4 What Scratch does

In this lesson

You will learn:

- about output.

Output comes out of the computer. The computer makes output. You can see or hear output.



Visual output

You can see some output. You see output on the computer screen.

Sound output

You can hear some output. You can listen on headphones. Or the sound might come out of the computer.



Activity

Press the space bar. The program starts. Scratch will chase the mouse pointer.

Let Scratch catch the mouse pointer.

What do you see on the screen?

What do you hear?



Think again

Draw and write to show the two outputs from this program.

Digital citizen of the future

Is it noisy in your classroom when lots of children use this program? Scratch has a loud meow. Sound output can be noisy. Listening on headphones means you don't disturb anyone else.



3.5 Think about input and output

In this lesson

You will learn:

➔ to think about input and output.

Input is what you do to control the computer.

Output is what comes out of the computer. The computer makes the output.

The Scratch program you used had input and output.



Activity

To make input you used:

- keyboard
- mouse.

Draw a picture to show the two types of input.



Activity

There are two outputs:

- visual output
- sound output.

Draw a picture to show the two types of output.



Extra challenge

Add word labels to your pictures. How can you control the program?

Explore more

Different types of computer have different inputs and outputs. Find out as many different kinds as you can.



Be creative

Some people have never used a computer. Think of a person like that. Write a letter to them. Tell them how you use a computer at school. Draw a picture to show them.

3.6 The Hedgehog Game

In this lesson

You will learn:

- to control the computer using inputs.



In this lesson you will play a **game**. It is called the Hedgehog Game. This game has two animals in it. They are Scratch the cat and a hedgehog.

- Scratch will follow the mouse pointer.
- The hedgehog will walk up and down.

A hedgehog is a spiky animal. If Scratch touches the hedgehog he will say "Ouch!".

Help Scratch to keep away from the hedgehog.



- Press any key to start the game.
- Press the space bar to make the hedgehog walk.
- Move the mouse so Scratch walks carefully.



Be creative

Choose a backdrop for the Hedgehog Game.



Think again

What are the inputs and outputs of this game? Draw or write to show them.

Extra challenge

Talk about other computer games you know. What were the inputs and outputs? Draw or write to show them.



Check what you know

You have learned

- ➔ to use the mouse to control the computer
- ➔ to use the keyboard to control the computer
- ➔ what inputs are
- ➔ to see and hear outputs.

Test

Here are some parts of a computer.

Mouse



Headphones



Screen



Keyboard



1. Point to a part that is used for input.
2. Draw or write any part that is used for output.
3. Draw a picture of a computer. Include all the parts used for input and output.

Activities



1 Play the Hedgehog Game (or any other game on a computer).

   Let your teacher see you work carefully at the computer.

2 Tell the teacher what you did.

3 Tell the teacher how you controlled the computer.

Self-evaluation

- I answered test question 1.
- I did activity 1 quietly and safely.
- I answered test questions 1 and 2.
- I did activities 1 and 2.
- I answered all the test questions.
- I did all the activities.

Re-read any parts of the unit you feel unsure about. Try the test and activities again – can you do more this time?

Digital citizen of the future

Computer games are good fun. But don't play a game when you are supposed to be working in class.

4 Programming: Play with Scratch

You will learn

- about blocks that make a program
- to make changes to a program.

In this unit you will look at a program made of blocks. You will learn how programs are made. You will make changes to the program.



Class activity

Scratch the cat is a **sprite**.



In the last unit you added a hedgehog sprite.



There are many other sprites. Here are some.



Which type of sprite do you like best?

Learning outcomes: Edit a program and say how that will change what it does

Digital citizen of the future

We often use programs that other people made. But we can also make changes. We can make programs do just what we want.

sprite block
program drag
right-click

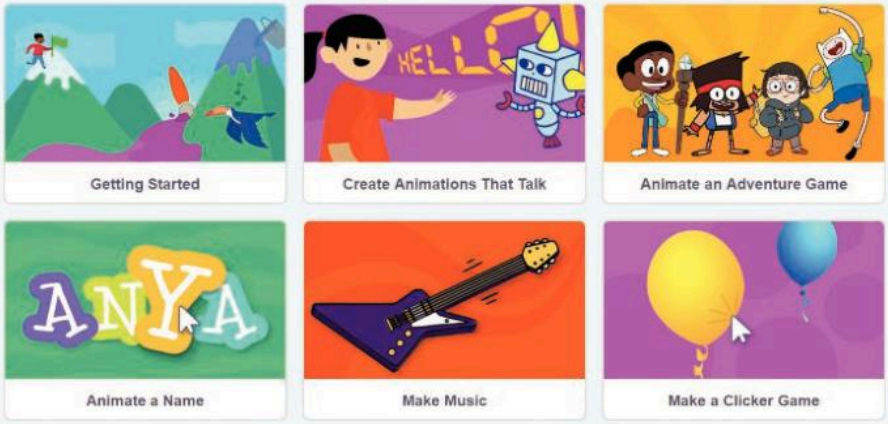
Be creative

Think of a new sprite that could be friends with Scratch. Draw the new sprite.



Did you know?

There are tutorials on the Scratch website. Look at the tutorials if you want to learn extra skills.



Talk about...

What computer games do you know? Which are your favourites?



4.1 Green flag means start

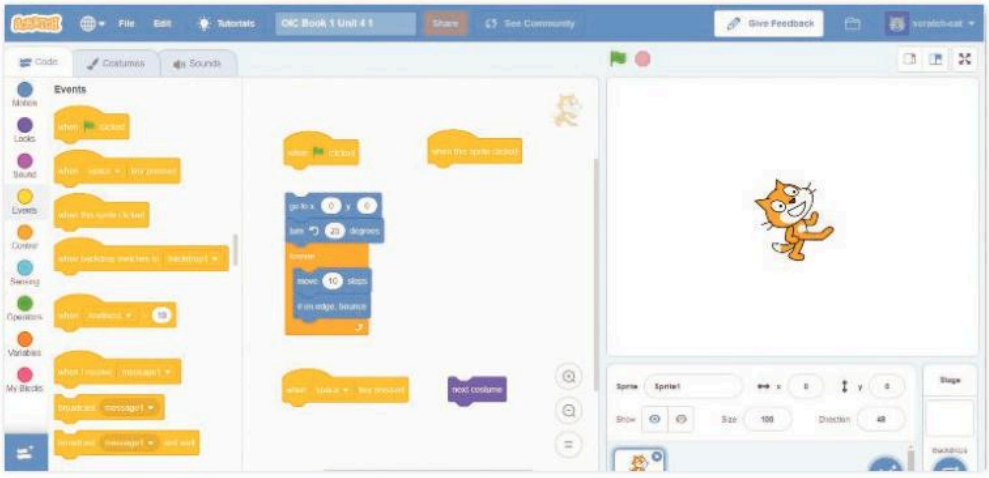
In this lesson

You will learn:

➔ to make a program start.

A **program** means some instructions. The instructions tell the computer what to do.

Look at this screen. In the middle of the screen is a program. The program is made of **blocks**. The blocks are shapes that fit together. Each block makes the computer do one thing.



This block tells the computer to start the program.

It has a green flag on it.



You can move the blocks and fit them together.

To move a block do this:

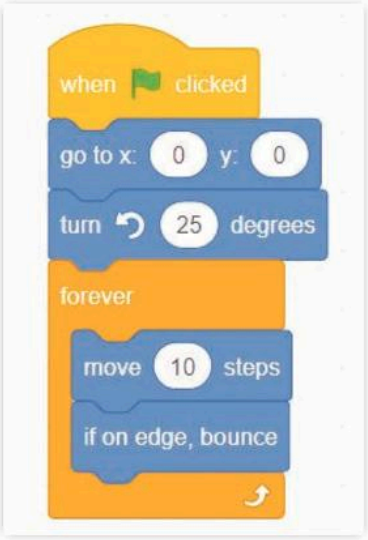
1. Move the mouse pointer to the block.
2. Press your finger on the mouse button. Keep pressing.
3. Move the mouse.

The block will move along too. This is called **dragging** the block.

When the block is in the right place, let go of the mouse pointer. The block will drop into the new place. This is called **drag and drop**.



Drag the green flag block to the program. When it touches the program the blocks will join together. It is like a jigsaw puzzle. Here is what the program will look like.



You joined the green flag block to the program.

Now click the green flag. The program will start. Scratch the cat will walk about.



Extra challenge

You learned how to add a backdrop to a program. Give this program a backdrop.



Think again

Explain what you did in this lesson. What change did you make? What happened?

4.2 What is a sprite?

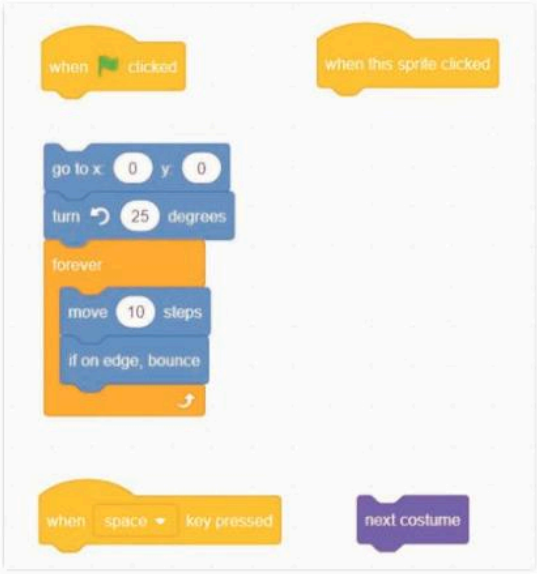
In this lesson

You will learn:

- how to make a program start in a new way
- what a sprite is.

A **sprite** is a picture on screen. A program makes the sprite move. Scratch is a sprite.

The program looks like this.



Some blocks have a curved top. A block with a curved top is a **start block**. A start block makes the program start.

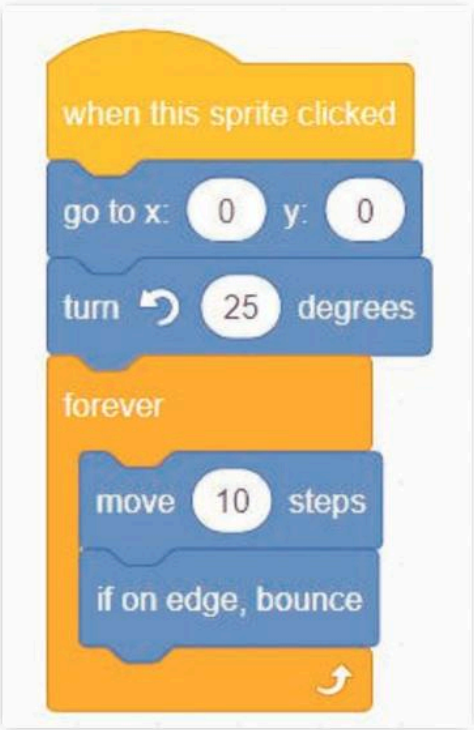
Here are two start blocks.



One has got the green flag. The other one says 'when this sprite clicked'.

Activity

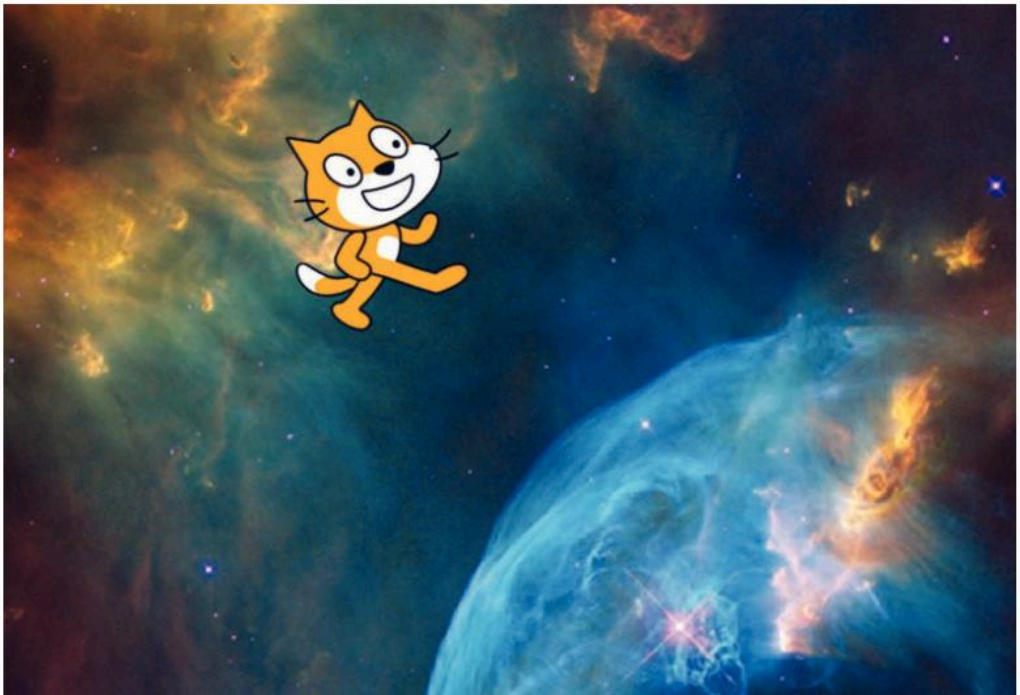
Join the second block to the program.



Click on the green flag. Nothing happens.

Click on the sprite. The program will start.

The sprite will move on the empty screen. Or on a backdrop if you chose one. This picture is an example.



Extra challenge

There is another start block on the screen. Can you find it? What happens if you use this start block?

Think again

What is a sprite? Say in your own words.

4.3 Costume change

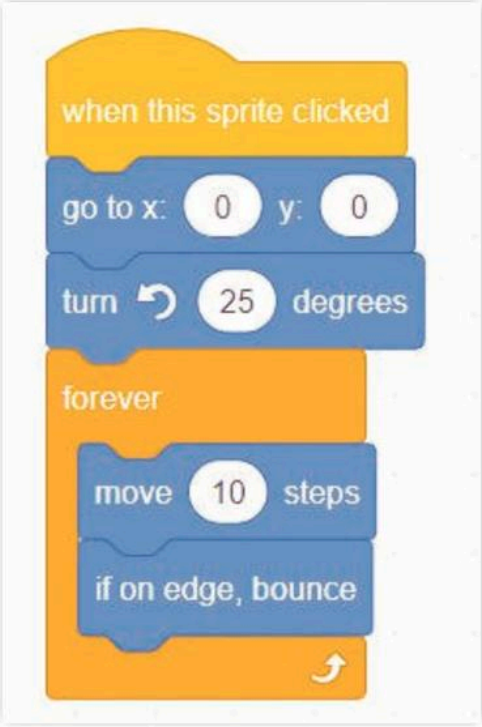
In this lesson

You will learn:

- to add a new block to the program.



Fit the program to a start block. You can choose which one.



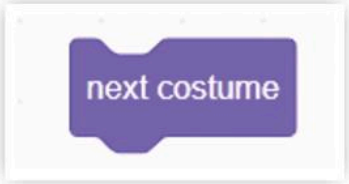
There is a purple block on the screen. Now you will add that block to the program.

Purple blocks change how Scratch looks. Scratch can swap between two looks. The looks are called **costumes**.





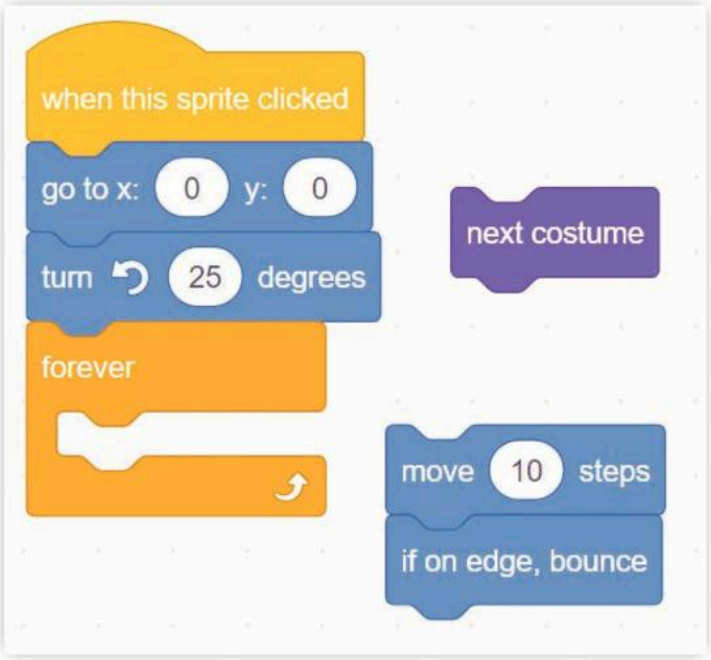
Changing costumes makes Scratch's legs move about.
This block says 'next costume'.

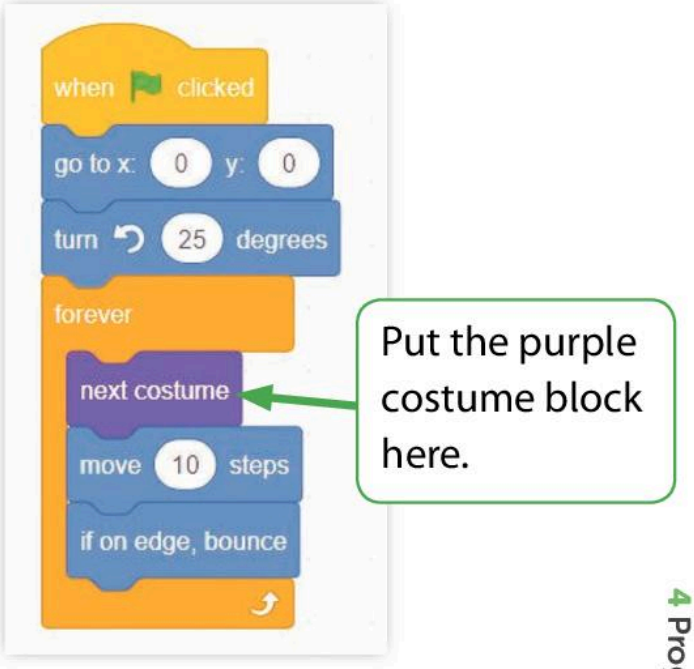


You will fit the purple costume block into the program.
First you must pull the blocks apart.

Then you can add the purple costume block. Make sure you put the it in the right place.

Then you can put the blocks back together again.





Make the program start. What is different?



Extra challenge

Where else can you fit the purple block? What happens?

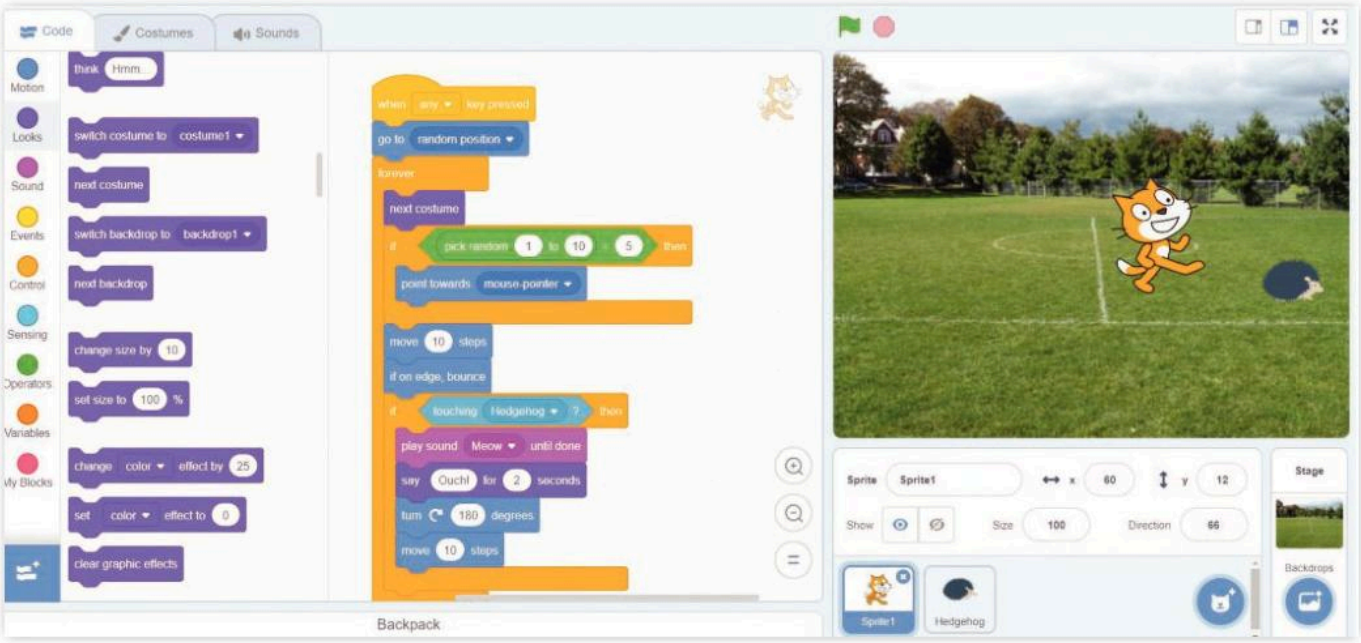
4.4 Extra sprites

In this lesson

You will learn:

➔ to add an extra sprite to a program.

You have seen this program before. It is called the Hedgehog Game. Scratch must dodge the hedgehog.



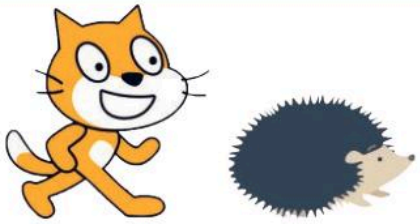
Scratch is a sprite. The program makes the sprite move.

Press the space bar to start the game.

There are two sprites in this game. Now you will add an extra hedgehog to the game.

Be creative

Draw a picture of Scratch and the hedgehogs.



Activity

You will add another hedgehog to the game.

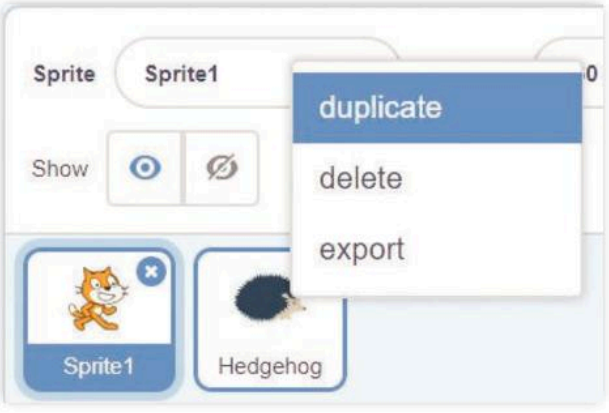
This part of the screen shows the sprites.



Move the pointer to the hedgehog.

Right-click the mouse. Right-click means to click the right button on the mouse.

You will see this menu.



Choose **duplicate**. Duplicate means to make two.

Click anywhere on the screen, then press the space bar to start the game. Now there are two hedgehogs.

Extra challenge

Can you add more hedgehogs to the game? Don't make it too hard for Scratch!

Think again

A classmate wants to make a copy of a sprite. Tell her how to do it.

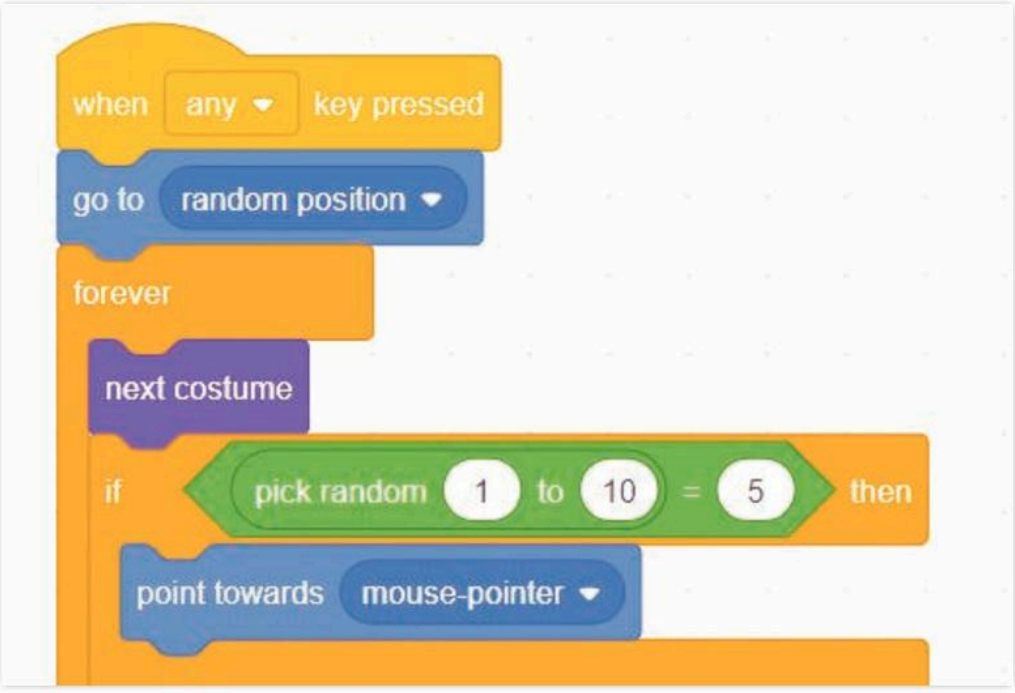
4.5 Find a new block

In this lesson

You will learn:

- how to find the block you want.

The program looks like this. The program will start when you press any key.



The first block is a start block.

Now you will change the start block. You will change the start block to the green flag block.

New blocks are stored on the left of the screen.

Extra challenge

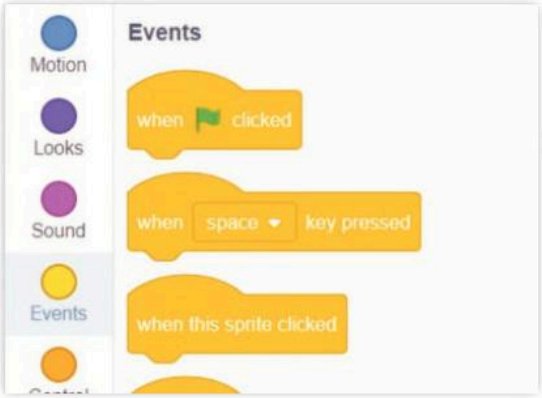
Click on the hedgehog sprite. Change the start block for the hedgehog.

Think again

Say the different ways you can start the game.

Activity

On the left are coloured dots. Click on the yellow dot.

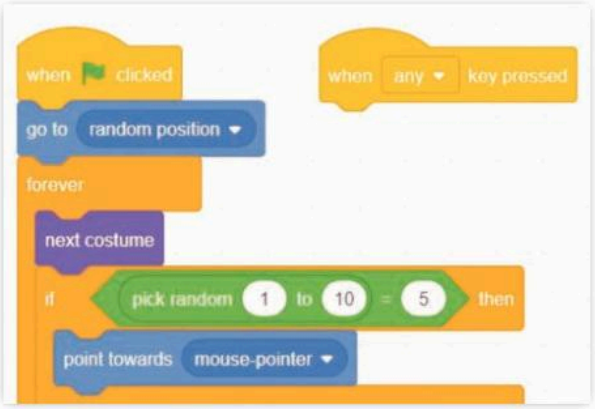


There are lots of start blocks. Find the green flag block.

Drag the block to the program. Drag means to hold down the mouse button and move the mouse.



Pull the program blocks away from the old start block. Fit the program to the green flag block.



Play the game. The green flag will start the game.